DATA STRUCTURE
LAB PROGRAM

15. Write a C program to implement hashing using Linear Probing method

```
#include <stdio.h>

#define SIZE 10

int hashTable[SIZE];

void initialize() {
    for (int i = 0; i < SIZE; i++)
        hashTable[i] = -1;
}

void insert(int key) {
    int index = key % SIZE;
    int originalIndex = index;

    while (hashTable[index] != -1) {
        index = (index + 1) % SIZE;

        if (index == originalIndex) {
            printf("Hash Table is Full\n");
            return;
        }
```

```
    }

    hashTable[index] = key;
}

void display() {
    printf("\nHash Table:\n");
    for (int i = 0; i < SIZE; i++) {
        if (hashTable[i] == -1)
            printf("%d --> Empty\n", i);
        else
            printf("%d --> %d\n", i, hashTable[i]);
    }
}

int main() {
    int n, key;

    initialize();

    printf("Enter number of keys: ");
    scanf("%d", &n);

    printf("Enter the keys:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &key);
        insert(key);
```

```
        }

        display();

        return 0;
    }
```

Output

```
Enter number of keys: 5
Enter the keys:
5 55 555 5555 55555

Hash Table:
0 --> Empty
1 --> Empty
2 --> Empty
3 --> Empty
4 --> Empty
5 --> 5
6 --> 55
7 --> 555
8 --> 5555
9 --> 55555


=== Code Execution Successful ===
```